Many of these examples are high stakes, like driving a car. You cannot just let a learning machine learn how to drive a car by trial and error. Fortunately, there are an increasing number of successful examples of letting learning machines loose in harmless simulators, and once they have mastered the simulator, letting them try real hardware in the real world. One instance that we will explore in this book is algorithmic trading. A substantial fraction of all stock trading is executed by computers with little to no input from human operators. Most of these algorithmic traders are wielded by huge hedge funds managing billions of dollars. In the last few years, however, we've seen more and more interest by individual traders in building trading algorithms. Indeed, Quantopian provides a platform where individual users can write trading algorithms in Python and test them in a safe, simulated environment. If the algorithms perform well, they can be used to trade real money. Many traders have achieved relative success with simple heuristics and rule-based algorithms. However, equity markets are dynamic and unpredictable, so a continuously learning RL algorithm has the advantage of being able to adapt to changing market conditions in real time.

One practical problem we'll tackle early in this book is advertisement placement. Many web businesses derive significant revenue from advertisements, and the revenue from ads is often tied to the number of clicks those ads can garner. There is a big incentive to place advertisements where they can maximize clicks. The only way to do this, however, is to use knowledge about the users to display the most appropriate ads. We generally don't know what characteristics of the user are related to the right ad choices, but we can employ RL techniques to make some headway. If we give an RL algorithm some potentially useful information about the user (what we would call the environment, or state of the environment) and tell it to maximize ad clicks, it will learn how to associate its input data to its objective, and it will eventually learn which ads will produce the most clicks from a particular user.

1.6. Why deep reinforcement learning?